

Modeling and Verification of Cross-Border Temporary Speed Restriction for High Speed Railway Train Control System Based on UPPAAL

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Abstract

Temporary speed restriction server (TSRS) is an important part of the high speed railway train control system, TSRS not only needs to check the temporary speed restriction orders issued by CTC, but also needs to exchange information frequently with temporary speed restriction server of adjacent dispatching station, on this account TSRS has higher requirements for security and real time property. In order to satisfy the real-time requirements, timed automata sub-models of each equipment are established for the working process of cross-border temporary speed restriction, and the overall timed automata network models are built through parallel composition of the sub-models. The BNF (backus-naur form) syntax is used to describe the properties of the cross-border temporary speed restriction server, and the verification tool of UPPAAL is used to simulate the models. The verification results confirm the safety and bounded liveness properties of the system in the interaction process.

Key Words: Cross-Border Temporary Speed Restriction, Timed Automata, UPPAAL, Real-time

1. Introduction

With the rapid development of Chinese high-speed railway, the requirements of the security and real-time in high speed railway train control system are becoming higher and higher. The temporary speed restriction server is an important part of the high speed railway train control system. Its working process has strict time constraints and security features and it's also a typical real-time system [1,2]. Therefore, according to the actual operation of high-speed trains, the temporary speed restriction may be set up in the CTCS (Chinese Train Control System) grade conversion zone, at this time, the one side of the CTCS grade conversion zone is the passenger dedicated line (CTCS-2/CTCS-3 level control

area), the other side is the existing speed control area [3–5]. Because of the complexity of information interaction between the devices, the formal methods have been used to verify the effectiveness of the system design defects and improve the security of the system.

The formal method is based on the discrete mathematics, and uses strict formal semantics to describe the characteristics of the system. Timed automata is one of formal methods, it can be used not only to describe the real value of the clock constraints system, but also to describe the state of the finite control variables of the system. Therefore, according to the CTCS grade conversion zone, the cross-border temporary speed restriction and the related equipments of information exchange process to establish the timed automata network models. So the UPPAAL (UPPAAL is an integrated tool environment for modeling, validation and verification of real-time sys-

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tems modeled as networks of timed automata, extended with data types) has been used to verify the safety and limited activity of cross-border temporary speed restriction.

2. Analysis of Temporary Speed Restriction in CTCS Grade Conversion Zone

In the high speed railway control system, the temporary speed restriction server not only carries on the verification and the storage to the temporary speed restriction order, but also must interact with other related signal equipments. Such as, CTC (centralized traffic control) is responsible to formulate, execute, and cancel the order. TSRS (temporary speed restriction server) completes calibration, distribution and synthesis, and feeds back to CTC. TCC (train control center), RBC (radio block center), GSM-R (global system for mobile communication-railway) send the TSR to vehicle equipment implementation. When the one side of the grade conver-

sion zone is the passenger dedicated line (CTCS-2/CTCS-3 level control area), the other side is the existing speed control area. TSRS not only needs to check the temporary speed restriction orders issued by CTC, but also needs to exchange information frequently with temporary speed restriction server of adjacent dispatching station [6,7]. The scene in the CTCS level conversion zone is shown in Figure 1.

The formulation of temporary speed restriction order follows the principle of “taking the forward line as the speed limit point”, speed restriction-2 on both sides of the TSRS data coverage. Therefore, TSRS-A and TCC-B have to achieve the synchronization of temporary speed restriction data interaction, so the temporary speed restriction orders are generated officially. This thesis is mainly aimed at the issued, cancellation and deletion of the temporary speed restriction orders [8,9], so as to verify the safety and bounded liveness properties of the system in the CTCS grade conversion zone.

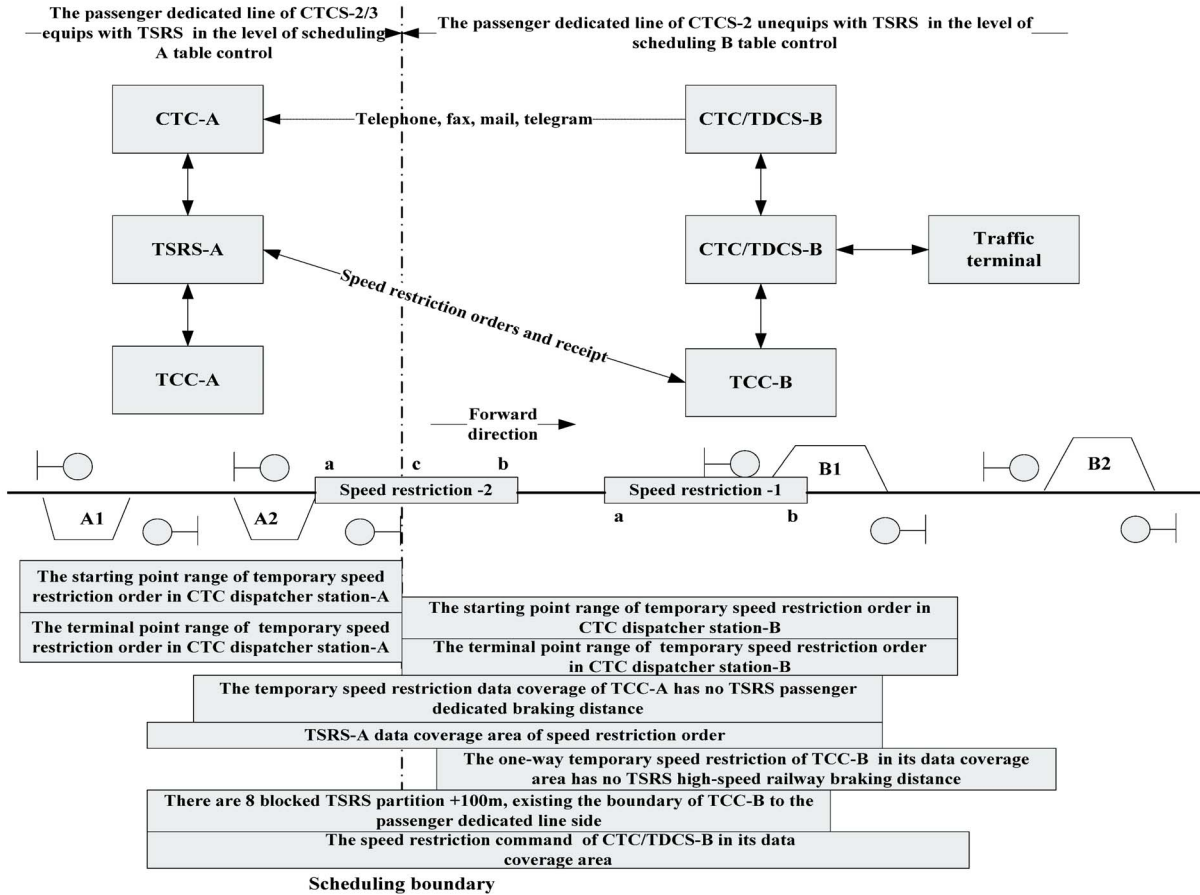


Figure 1. Temporary speed restriction scene in CTCS grade transition zone.

2.1 The Formulation of Cross-border Temporary Speed Restriction Order

As shown in Figure 1, the speed restriction-2 is responsible for the development of a-c section by CTC-A, CTC/TDCS-B is responsible for the development of c-bsection. If the left side of the scheduling platform CTC-A receives the temporary speed restriction orders from TSRS-A, meanwhile the temporary speed restriction command will be checked by TSRS-A, the verification process is shown in Figure 2. In addition, TSRS-A receives the temporary speed restriction orders from TCC-A and RBC-A, and meanwhile the verification and execution of the orders have been completed in the right side of the TCC-B. TCC-B will validate the execution results and feed back to TSRS-A with its comprehensive validation results.

If the temporary speed restriction orders are issued to TCC-B on the right side of TDCS-B, and then TCC-B sends the speed restriction orders to TSRS-A on the left side (the flow chart is shown in Figure 3). If the inspection is successful, the orders should be sent to the RBC-A and TCC-A for executing. If the inspection fails, TSRS-A will perform the checking results of failure to dispatcher station on the right side of CTC/TDCS-B. The

information of comprehensive verification will be fed back to the TSRS-A, the results are sent to the left side of the dispatcher station.

2.2 Cancellation and Deletion of Cross-border Temporary Speed Restriction Order

The speed restriction orders can only be revoked by the CTC-A in the passenger dedicated line. The information of the existing speed line dispatcher station TDCS needs to be confirmed by telephone or fax and sends to the passenger dedicated line station. After the confirmation, the order of canceling can be issued by the TDCS [10].

3. Modeling and Simulation of the System

3.1 The Timed Automata and the UPPAAL

The theory of timed automata was proposed by R. Alur and Dill in 1994, timed automata is a kind of formal method for modeling, which uses clock variables to represent the time and uses the associated time constraints to represent the state transition diagrams. The validation tool of UPPAAL was developed by Aalborg University and Uppsala University in 1995, which uses

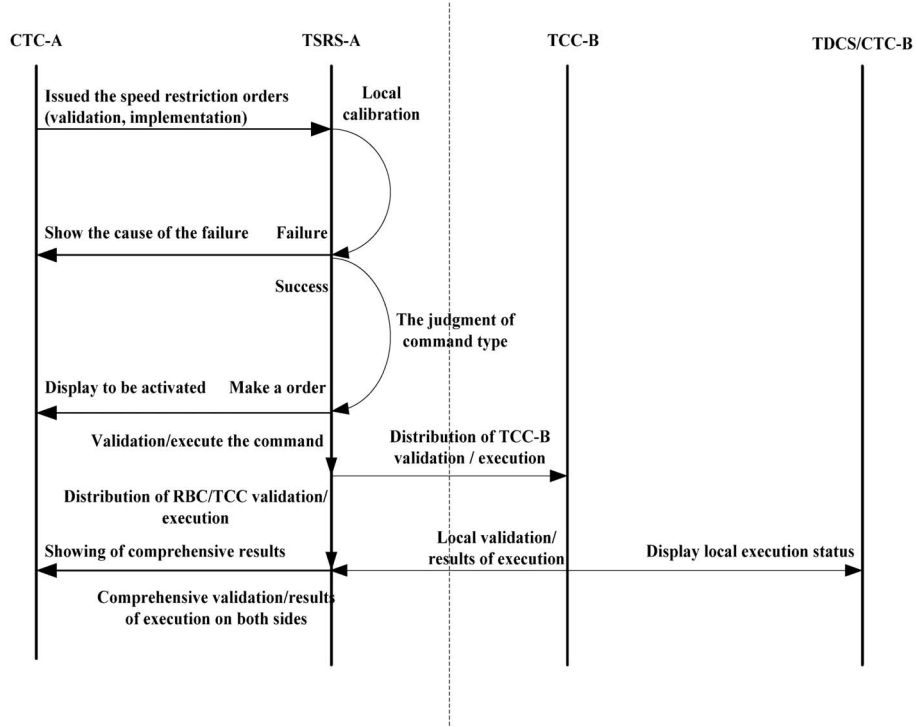


Figure 2. The process of calibration in passenger dedicated side.

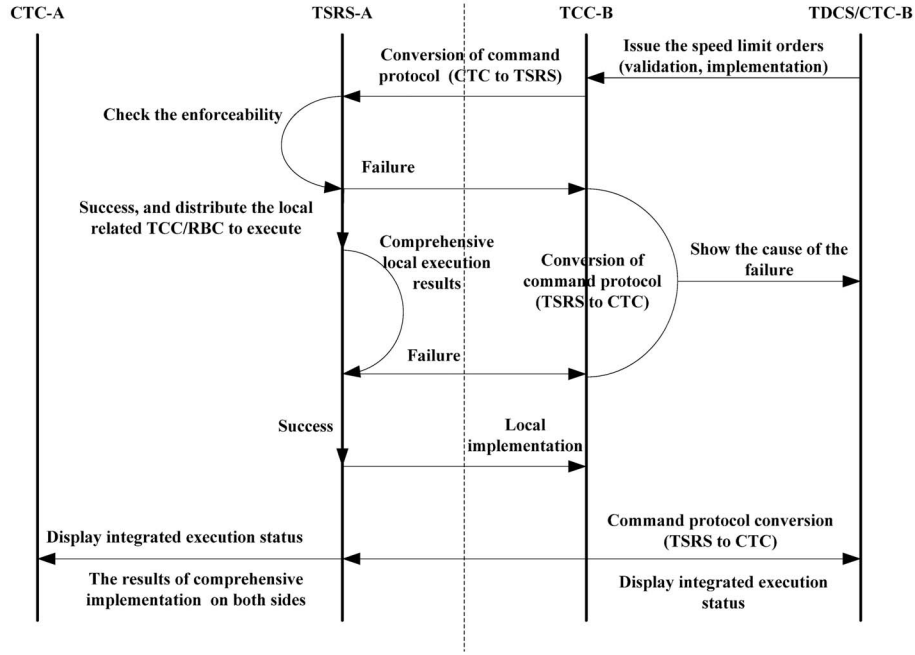


Figure 3. The process of calibration in the existing side.

integer variable to simulate the product of real time system. By means of a finite number of control structures and numerical clock variables, it verifies the security and bounded liveness of the system.

The UPPAAL user interface includes three parts: system editor, simulator and verifier. Among them, the system editor is used to create and edit the system with analysis. The simulator is used to check whether the system models are correct or not. The verifier is used to search the system models of the state space with checking the clock constraints and restrictions on the properties [11–14]. The validation tool of UPPAAL applies the

BNF syntax: $\text{Prop} ::= E <> p \mid E[]p \mid A <> p \mid A[]p \mid p \rightarrow q$, this grammatical meaning is shown in Table 1.

3.2 Based on Timed Automata Models of the Cross-border Temporary Speed Restriction

The modeling and simulation tool of UPPAAL is used to verify the working process of the 7 moving objects of the cross-border temporary speed restriction information exchange in the CTCS grade transition area, namely retransmission controller, clock controller, queue controller, CTC-A, TSRS-A, TCC-B, CTC-B. Each timed automata model is constructed, as is shown in Figure 4.

Table 1. BNF syntax and its implications

BNF syntax	Specific meaning
$E <> p$ indicates that the property of p may be satisfied	$E <> p$ is true, if and only if there is a sequence in the conversion system $S_0 \rightarrow S_1 \rightarrow \dots \rightarrow S_n$, making S_0 is the starting state, S_n is p .
$E[]p$ indicates that the property of p is always possible	$E[]p$ is true, if and only if there is a sequence $S_0 \rightarrow S_1 \rightarrow \dots \rightarrow S_i \rightarrow \dots$, it makes p in all state.
$A <> p$ indicates that the property of p is eventually satisfied	$A <> p$ is true, it indicates that all possible path state sequences will eventually go through a state, this state satisfies the property of p .
$A[]p$ indicates invariantly	$A[]p$ indicates that the property of p is invariant, it is equivalent to $\text{no } tE <> \text{not } p$.
$p \rightarrow q$ indicates leading to	$p \rightarrow q$ indicates that the property of p satisfies to lead to the property of q and it is equivalent to $A[] (p \text{ imply } A <> q)$.

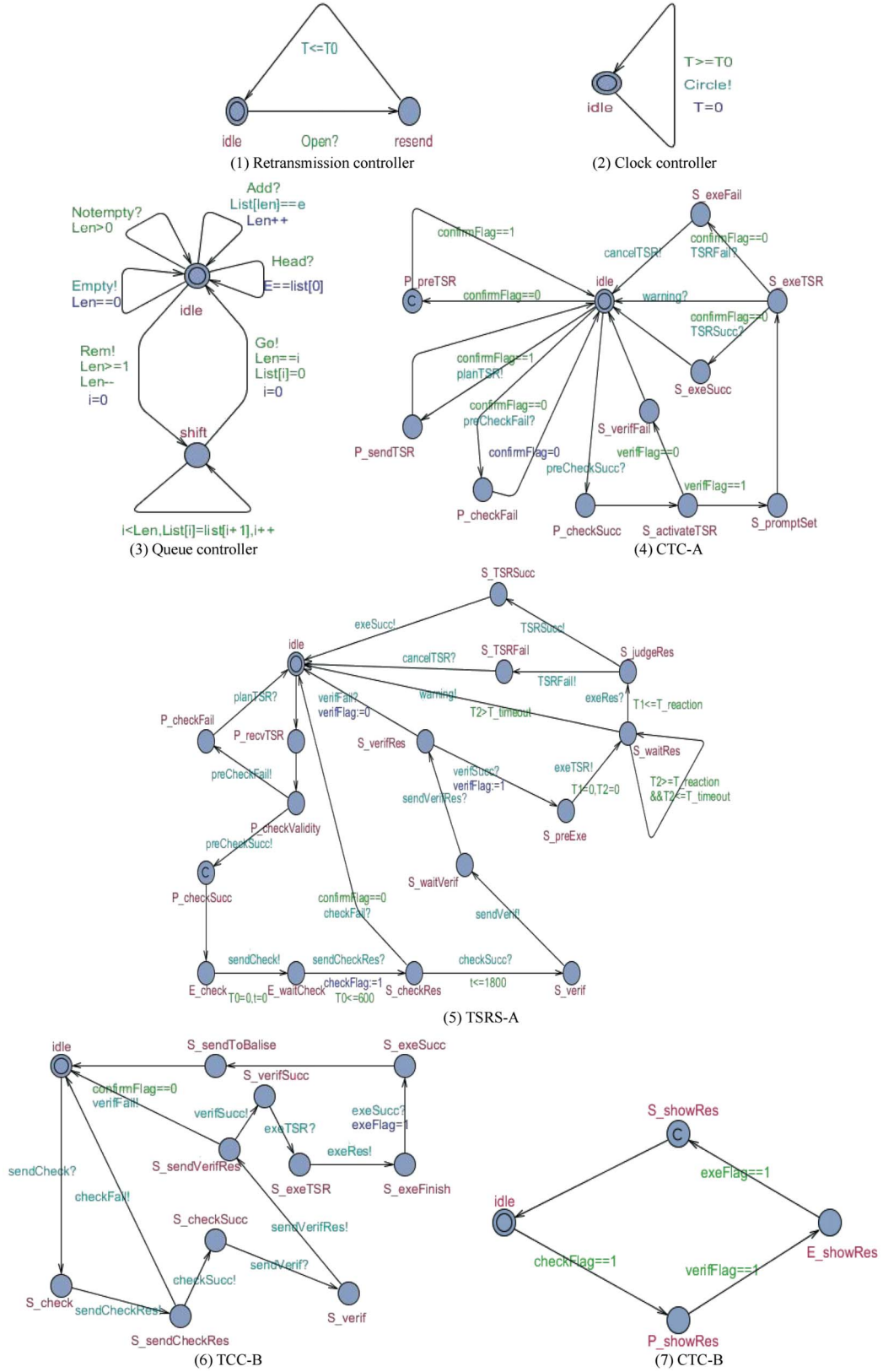


Figure 4. Timed automata models of the cross-border temporary speed restriction information exchange in the CTCS grade transition area.

Settimed automata models of the cross-border temporary speed restriction in CTCS grade transition zone:

$$TA = \langle S, S^0, A, X, I, E \rangle$$

S, S^0, A, X, I, E represent the process state for modeling of time automata respectively, Initial state, Trigger event, Time parameter, Integer variable, State transition path. Among them, the state of the process $S = \{\text{idle}, P_preTSR, S_exeFail, S_exeTSR, \dots\}$. Initial state $S^0 = \text{idle}$. Trigger event $A = \{\text{cancelTSR}, \text{TSRFail}, \text{warning}, \text{planTSR}, \dots\}$. Time parameter $X = \{T0, T1, T2, t, T_TCCreaction, T_TSRStimeout\}$, where $T0$ indicates a temporary speed restriction command prompting time, $T1$ and $T2$ are used to represent the communication time between TSRS, TCC and CTC, $T_TCCreaction$ represents the waiting time for the TCC to receive the message and feed back the message to TSRS, $T_TSRStimeout$ indicates that the received feedback information is timed out. Integer variable $I = \{\text{confirmFlag}, \text{verifFlag}, \text{checkFlag}, \text{exeFlag}\}$. State transition path $E = \{\text{idle}, \text{comfirmFlag}, P_preTSR, \text{confirmFlag}, \text{idle}, \dots\}$. Where the “P_”, “E_”, “S_” at the beginning of the speed restriction indicate the development of the ordering, is-

sued and setting the process respectively.

As is shown in Figure 4, the timed automata models of the cross-border temporary speed restriction information. Among them, “!” represents the issue of information exchange, “?” represents the receiving of information exchange. A circle marked with C is a firm position, it represents an immediate transition from the constraint position without a time delay. The main position in the models are shown in Table 2.

4. Validation of the System Model

In the UPPAAL the timed automata network models of the cross-border temporary speed restriction have been established by editor in CTCS grade transition area, whose working process is shown in Figure 5. At the same time, in the simulator, the cross-border temporary speed restriction orders are issued, canceled and deleted, and the interaction of information is simulated and checked. Finally in the UPPAAL validator, the BNF syntax is used to describe the properties of the cross-border temporary speed restriction server and confirm the safety and bounded liveness properties of the system in the interaction process. As is shown in Figure 6.

Table 2. Main locations in the model

CTC-A	TSRS-A	TCC-B	TSRS-B
P_preTSR: Prepare TSR	P_recvTSR: Receive TSR	S_sendCheckRes: Send the commands of checking	P_showRes: The show of comprehensive proposed results
P_sendTSR: Send TSR	P_checkValidity: The validation checking of TSR	S_sendVerifRes: Send validation commands of execution	E_showRes: The show of comprehensive released results
S_activateTSR: Activate TSR	E_check: Checking	S_exeTSR: Exceute TSR	S_showRes: The show of comprehensive implementation results
S_promptSet: Prompt setting	E_waitCheck: Waiting for the results of verification	S_exeFinish: The results of execution	
S_exeTSR: Issued the implementation of TSR	S_checkRes: The results of verification for both parties	S_sendToBalise: Send to balise	
P_checkFail: Check fail	S_verif: Verification		
P_checkSucc: Check success	S_preExe: Pre execution		
S_exeFail: Execute fail	S_waitRes: Waiting for the results		
S_exeSucc: Execute success	S_judgeRes: Judging the executing results		

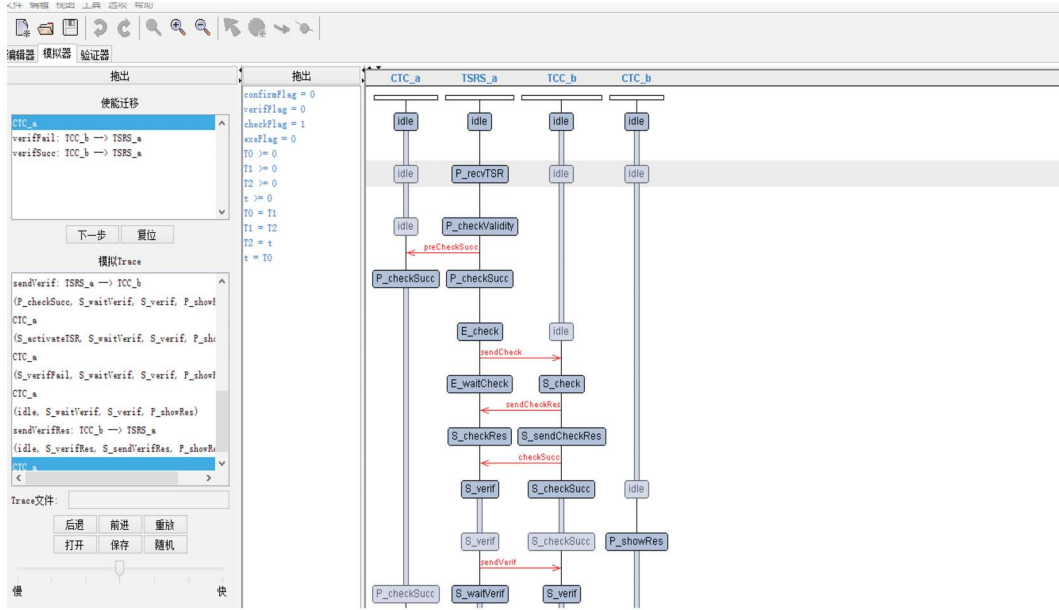


Figure 5. The working status and diagram of cross-border temporary speed restriction.

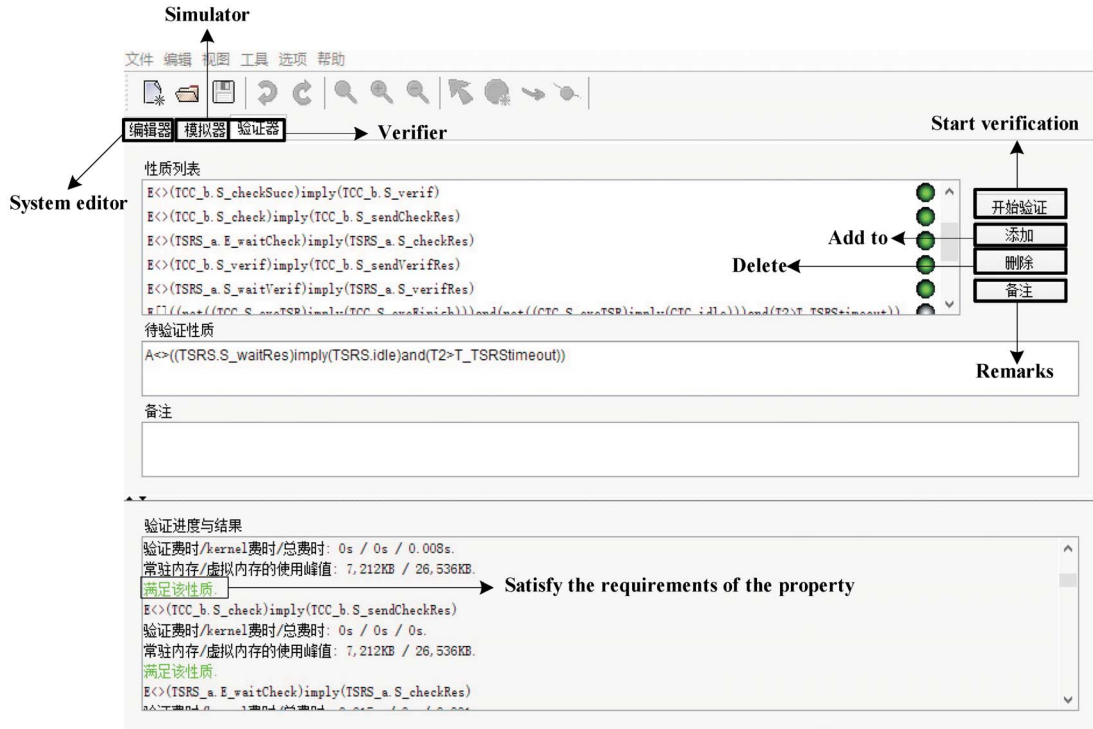


Figure 6. Properties validation.

4.1 The Requirements of Security in the System

(1) The temporary speed restriction command distributed by TSRS-A should be verified, meanwhile the adjacent station TCC has to check the command.

$E < > ((TSRS_a.idle) \text{ imply } (TSRS_a.E_waitCheck) \text{ and } (TCC_b.idle) \text{ imply } (TCC_b.S_check))$. Validation is passed.

(2) The implementation of the temporary speed restric-

tion orders has been performed in TSRS-A and the adjacent station TCC has been requested to verify the validity of the orders.

$E < > ((\text{TSRS_a.idle}) \text{ imply } (\text{TSRS_a.S_verif}) \text{ and } (\text{TCC_b.S_checkSucc}) \text{ imply } (\text{TCC_b.S_verif}))$. Validation is passed.

- (3) TCC should feed back and distribute the checking results to the TSRS-A.

$E < > ((\text{TCC_b.S_check}) \text{ imply } (\text{TCC_b.S_sendCheckRes}) \text{ and } (\text{TSRS_a.E_waitCheck}) \text{ imply } (\text{TSRS_a.S_checkRes}))$. Validation is passed.

- (4) TCC should feed back the results of verification to the TSRS-A and execute the orders.

$E < > ((\text{TCC_b.S_verif}) \text{ imply } (\text{TCC_b.S_sendVerifRes}) \text{ and } (\text{TSRS_a.S_waitVerif}) \text{ imply } (\text{TSRS_a.S_verifRes}))$. Validation is passed.

- (5) Neither does TCC-B send the results of implementation to TSRS-A, TSRS-A also does not alarm to the CTC-B.

$E[] ((\text{not } ((\text{TCC_b.S_exeTSR}) \text{ imply } (\text{TCC_b.S_exeFinish}))) \text{ and } (\text{not } ((\text{CTC_b.S_exeTSR}) \text{ imply } (\text{CTC_b.idle})))) \text{ and } (\text{T2} > \text{T_TSRStimeout}))$. Validation is not passed, because no feedback and no alarming of information is allowed in this situation.

4.2 The Requirements of Bounded Liveness in the System

- (6) The temporary speed restriction orders have been received by TCC-B, the results of the verification requirements should be fed back to TSRS-A in the 1s.

$A < > (\text{TCC_b.S_exeTSR}) \text{ imply } (\text{TCC_b.S_exeFinish}) \text{ and } (\text{T1} \leq 1)$. Validation is passed.

- (7) The validity of the speed restriction information has been verified by TSRS-A. The calibration results should be fed back to CTC-A station in 2s. If the verification is successful, the temporary speed restriction for the operation of the terminal has to prompt the activation of the temporary speed restriction command to CTC-A in every 10 min.

$A < > ((\text{TSRS_a.P_checkValidity}) \text{ imply } (\text{TSRS_a.P_checkSucc}) \text{ or } (\text{TSRS_a.P_checkValidity}) \text{ imply } (\text{TSRS_a.P_checkFail}) \text{ and } (\text{T2} < 2) \text{ and } (\text{CTC_a.S_judgeCheckRes}) \text{ imply } (\text{CTC_a.S_waitConfirm}) \text{ and } (\text{T0} = 600))$. Validation is passed.

- (8) If TSRS-A does not receive the feedback informa-

tion in excess of T_TSRStimeout time, it should alarm.

$A < > ((\text{TSRS_a.S_waitRes}) \text{ imply } (\text{TSRS_a.idle}) \text{ and } (\text{T2} > \text{T_TSRStimeout}))$. Validation is passed.

Each of the above properties are verified, indicating that the system satisfies the requirements of the safety and bounded liveness of the cross-border temporary speed restriction for the CTCS grade transition area.

5. Conclusions

This paper applies the formal method of timed automata for modeling, and establishes the network of timed automata models in the whole information interactive system, meanwhile the validation of the models is carried out by using UPPAAL. The safety and bounded liveness of the cross-border temporary speed restriction in the CTCS grade transition zone are verified successfully, and that ensures the smooth running of the system with providing an important basis for the further development of the system.

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